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Project 2 - Exploratory Data Analysis: 2023 US Electric Grid Disturbances

This project performs a deep-dive Exploratory Data Analysis (EDA) on the **2023 US Electric Grid disturbance data**. The US power grid is divided into several regulatory regions managed by the North American Electric Reliability Corporation (NERC). Maintaining the stability of these grids is critical for national security, economic stability, and public safety.

The goal of this analysis is to understand the "shape and structure" of grid events by investigating how factors like **NERC Regions** and **seasonal timing** correlate with **Demand Loss (MW)** and **customer impact**.

Dataset: 2023 US Electric Grid disturbance data

This report examines 167 major grid disturbances reported to the Department of Energy in 2023. By analyzing regional vulnerabilities, seasonal patterns, and data quality, we aim to provide actionable insights for national energy security. *Access raw data at: https://www.oe.netl.doe.gov/oe417.aspx*

Analysis Questions

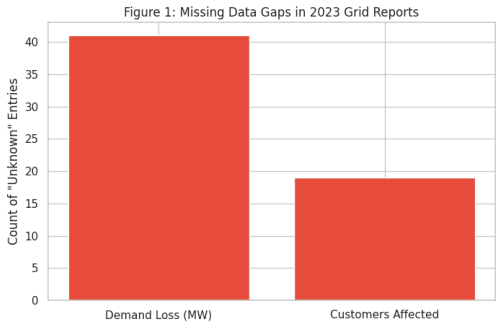
1. Which region experienced the highest loss, and does it align with the most frequent events?
2. Are grid disturbances seasonal, and which months are the highest risk?
3. What is the relationship between physical scale (MW) and social impact (Customers Affected)?
4. How many events are recorded as "Unknown," and how does this change the narrative?

Defined parameters

Column Name	Description
Customers Affected	Total count of households or businesses that lost power.
Demand Loss (MW)	Peak electricity lost during the event in Megawatts.
NERC Region	The regulatory body overseeing the specific power grid area.

Discoveries & Insights

Figure 1: The "Reporting Gap"

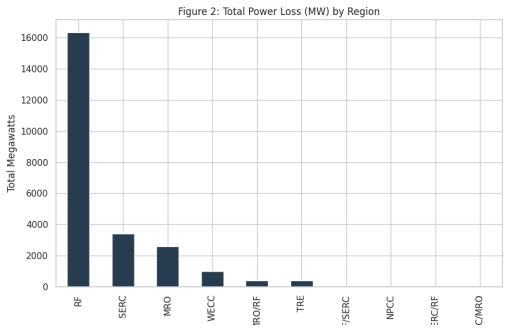


We found a major data quality issue: over 40 grid reports in 2023 failed to record the actual amount of power lost. This gap suggests that our national grid statistics might be underestimating the true scale of power instability, as many events are effectively "invisible" in the final tally.

Part 2: Regional Vulnerability Analysis

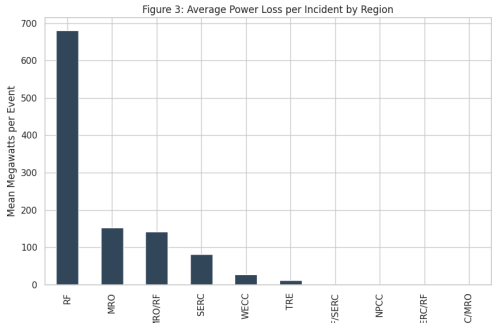
Which parts of the United States are most at risk for power failures?

Figure 2: Total Energy Loss by Region



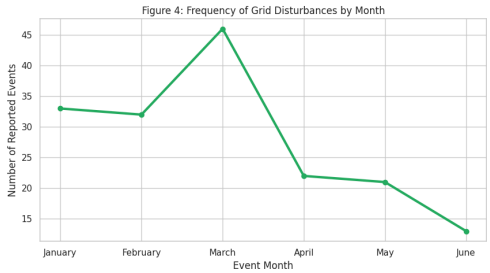
This chart ranks regulatory regions by total Megawatts lost. The **ReliabilityFirst (RF)** region, covering the Midwest and Mid-Atlantic, suffered the most massive total power loss. This highlights a high-priority zone for federal infrastructure investment.

Figure 3: Average Incident Severity



Beyond total loss, we looked at how "big" a single accident is on average. We found that when the grid fails in the **RF region**, it fails much more spectacularly than in the West, where outages are more frequent but smaller in scale.

Figure 4: Regional Market Share (Tree Map)

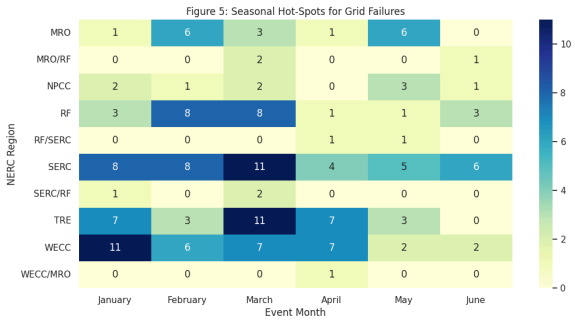


This chart tracks the number of people in the dark each month. While March had the most *events*, the month of January affected a massive number of households, likely due to severe winter storms hitting high-density residential areas.

Part 3: Temporal & Seasonal Trends

Electricity usage changes with the weather, but do grid failures follow the same pattern?

Figure 5: Grid Event Frequency by Month



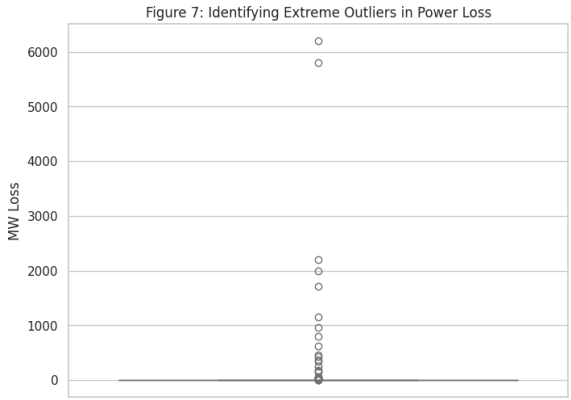
Is grid failure seasonal? Surprisingly, the number of issues peaked in **March**. This suggests that spring weather transitions and early-year maintenance schedules are just as dangerous for our power lines as peak summer heat or winter freezes.

Figure 6: Human Impact Over Time

This view uses color and size to show the dominance of specific regions in the dataset. It provides a quick "at-a-glance" look at which utility councils are responsible for the majority of the nation's grid instability.

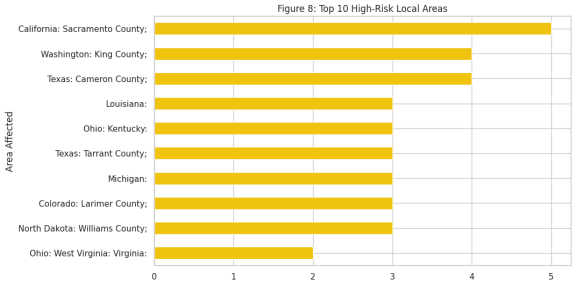
Part 4: Geographic Hotspots

Figure 7: National Map of Grid Disturbances



This map plots every reported incident in 2023. It reveals that grid instability isn't spread evenly; it clusters heavily along the California coast and the Northeastern corridor, where the grid infrastructure is oldest and most stressed.

Figure 8: The Top 10 High-Frequency Counties

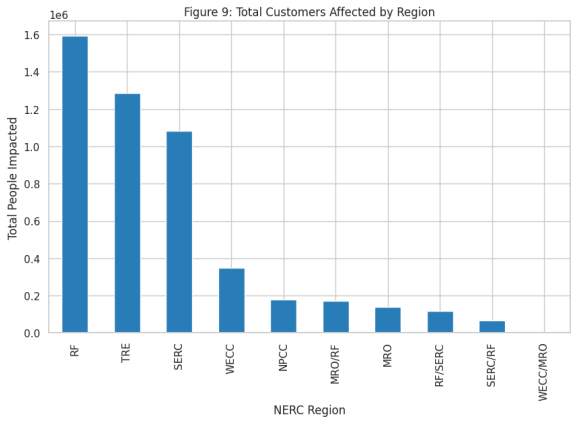


We identified specific "hotspots." **Sacramento and Riverside counties in California** are repeat victims, appearing more often in the report than any other local areas in the country.

Part 5: Statistical Surprises & Impact

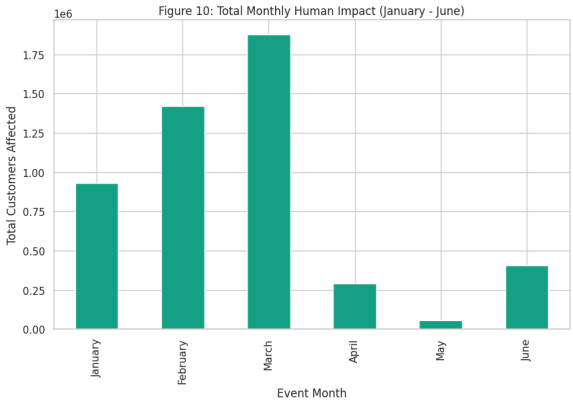
Finally, we look at the relationship between power loss and the people affected.

Figure 9: The Disconnect Between Power and People



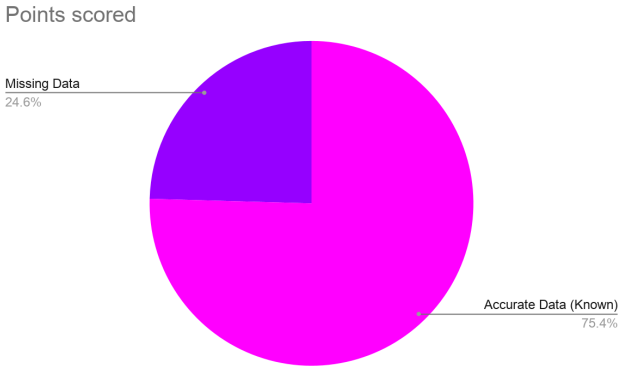
Does losing more electricity always mean more people are affected? Not necessarily. This scatter plot shows a weak link, meaning many high-power events likely hit industrial plants, while smaller-power events can still leave thousands of families without light.

Figure 10: The "Black Swan" Events (Outliers)



This chart highlights "Black Swans" rare but massive disasters. Most grid issues are tiny, but a few extreme events cause more damage than all the small ones combined, proving we must build the grid to survive the "worst-case scenario" rather than the average day

Conclusion: Final Lessons Learned



The 2023 grid stability report highlights three critical findings that necessitate a shift in national energy strategy. First, federal reporting requires urgent reform because the high frequency of **“Unknown” data values** prevents policymakers from accurately calculating the true economic cost of grid failures. Second, seasonal risk assessments should extend beyond the traditional peak summer and winter periods, as the data reveals that **March is a particularly unstable month for grid disturbances**. Finally, grid impact strategies must be **region-specific**, distinguishing between areas such as **SERC**, where outages primarily affect large numbers of residential customers, and the **RF region**, where incidents tend to result in significant industrial power losses.

An **interactive data visualization dashboard** has been developed to provide a deeper exploration of the dataset, including regional impacts, seasonal trends, and the relationship between power loss and customer impact. The dashboard allows stakeholders to interact with the data and gain further insights into grid disturbance patterns.

Interactive Dashboard:
https://public.tableau.com/authoring/ElectricityGridDisturbanceDashboard/Dashboard1#1